

NeuroFleetX: Intelligent Taxi Dispatch System

This report details the backend development, AI integration, and machine learning model integration for NeuroFleetX, an intelligent taxi dispatch and fleet management system. It showcases the creation of a scalable platform designed for efficient ride, user, and driver management, enhanced by advanced AI and ML capabilities.

Project Overview & Objectives

NeuroFleetX aims to establish a robust and scalable backend platform capable of handling complex operations while integrating cutting-edge AI and ML functionalities. The core objectives centered around efficient management and intelligent prediction services.

Secure Management

Establish secure user, driver, and admin management systems.

Efficient Ride Handling

Streamline ride request processing and real-time tracking.

Intelligent ETA Prediction

Integrate an existing ML model for accurate estimated time of arrival predictions.

AI-Powered Insights

Provide AI-powered explanations for ride history for enhanced transparency.

Seamless Communication

Ensure robust frontend-backend communication channels.

My Role and Core Responsibilities

As a Backend Developer, my primary focus was on implementing the complete backend system for NeuroFleetX, excluding taxi-specific APIs. This involved a broad spectrum of development tasks ,from architectural design to integration of intelligent services.



Backend Architecture

Designed the system using Spring Boot.



RESTful API Development

Developed and secured all necessary APIs.



Authentication & Authorization

Implemented JWT-based security protocols.



Database Management

Managed schema and relationships with JPA.



AI/ML Integration

Connected pre-existing ML model and OpenAI API.



Important Clarification: I focused on integrating an existing ETA prediction ML model; I was not responsible for its development or training.

Backend Architecture & Technology Stack

The NeuroFleetX backend adheres to a robust, layered architecture, ensuring modularity, scalability, and ease of maintenance. This structure facilitates clear separation of concerns and efficient development.

Layered Architecture

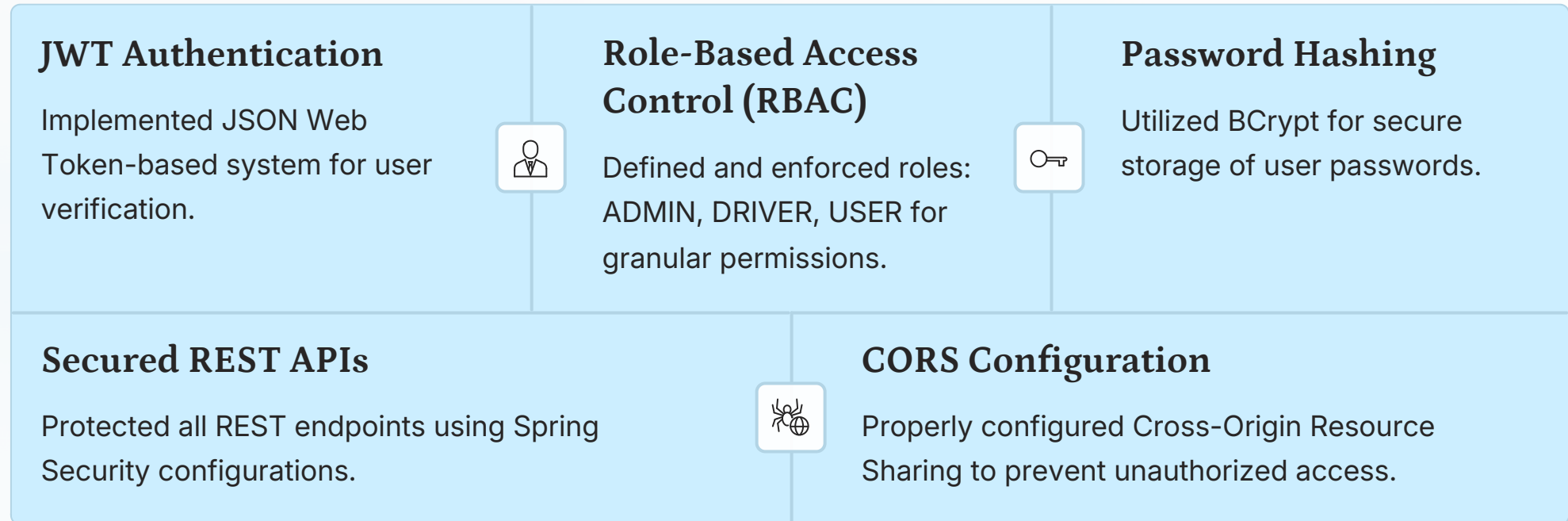
- **Controller Layer:** Handles HTTP requests and responses.
- **Service Layer:** Encapsulates core business logic.
- **Repository Layer:** Manages database interactions via JPA.
- **Model Layer:** Defines entities and data structures.

Key Technologies

- **Backend:** Java 17, Spring Boot 3, Spring Security, Spring Data JPA & Hibernate, PostgreSQL, JWT Authentication.
- **AI & ML:** OpenAI API, Pre-trained ML ETA prediction model.
- **Frontend Integration:** React, Axios, Tailwind CSS.

Robust Security Implementation

Security was a paramount concern in the development of NeuroFleetX. A comprehensive suite of measures was implemented to protect data and ensure secure access to system resources.



Comprehensive Backend API Implementation

A diverse set of RESTful APIs was developed to support all functionalities of the NeuroFleetX system, ensuring seamless interaction between the frontend and backend, with strict adherence to security and role-based access .

1

Authentication APIs

- Login, Signup, Logout
- Get current user details

2

User APIs

- Profile management
- Ride history access

3

Driver APIs

- Ride acceptance/rejection
- Earnings tracking

4

Admin APIs

- User management
- Ride analytics
- ML model usage & metrics

All APIs were designed to be REST standards-compliant and secured with role-based access controls.

Database Design and Management

The foundational data layer of NeuroFleetX was built on PostgreSQL, chosen for its robustness and reliability. My contribution included designing the relational schema and ensuring data integrity and consistency throughout the system.



PostgreSQL

Selected as the primary database.



Schema Design

Created an optimized relational database schema.



JPA & Hibernate

Implemented entity relationships and managed schema updates.

Key entities modeled included **User**, **Driver**, **Ride Request**, **Taxi** (structure only), and **ML-related data**.



Data Consistency

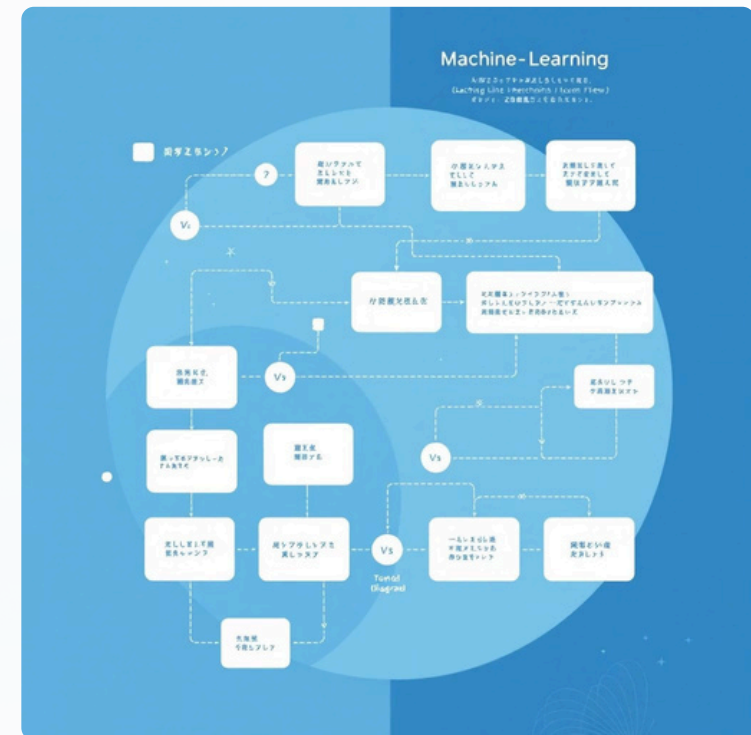
Ensured high levels of data integrity.

Machine Learning Model Integration

A crucial aspect of NeuroFleetX's intelligence lies in its ETA prediction capabilities. My role involved integrating an existing machine learning model into the backend, enabling dynamic and accurate arrival time estimations.

Seamless ML Integration

- **Backend Integration:** Successfully integrated the pre-trained ML model with core backend services.
- **API Development:** Created dedicated APIs to feed required input features to the model.
- **Frontend Delivery:** Ensured efficient transmission of ETA predictions back to the user interface.
- **Data Flow Optimization:** Guaranteed smooth and reliable data flow between frontend, backend, and the ML model.



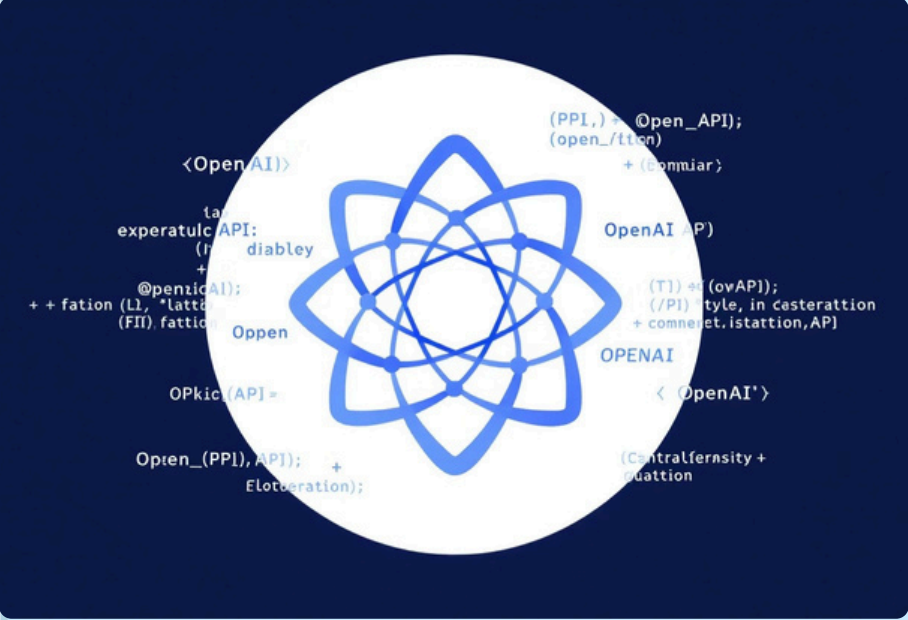
The ML model is used solely as an integrated service to predict estimated arrival times based on ride parameters; no training or algorithm design was part of my responsibilities.

AI Integration with OpenAI API & Frontend Interoperability

One of my key contributions was the integration of AI-powered ride history summarization using the OpenAI API, significantly enhancing the system's intelligence and user experience. Alongside this, robust frontend interoperability was ensured.

OpenAI API Integration

Leveraged the OpenAI API to analyze raw ride data.



Intelligent Summarization

Generated human-readable explanations of complex ride details, boosting transparency for users and admins.



Frontend Compatibility

Designed clear API contracts and enabled secure token-based communication for seamless frontend integration.



UI Verification

Verified correct role-based behavior and data display within the frontend user interface.



Key Learnings & Conclusion

The NeuroFleetX project was an invaluable experience, providing hands-on exposure to advanced backend development and intelligent service integration. It solidified my understanding of building scalable, secure, and smart applications.

1 Enterprise Backend Development

Mastered Spring Boot for robust backend solutions.

2 Advanced Security Practices

Implemented secure authentication and RBAC with JWT.

3 AI & ML Integration

Successfully integrated OpenAI API and an existing ML model into a production-ready system.

4 Database & Deployment

Gained practical experience in database design and deployment strategies using Docker.

This project showcased my ability to develop a full-featured backend, connect disparate intelligent services, and contribute to a real-world application with high standards of security and performance.

